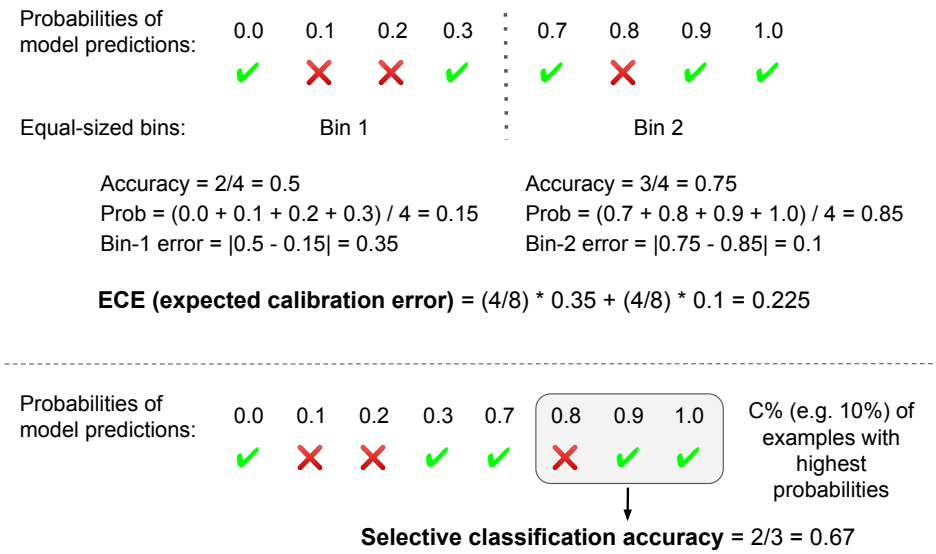

Figure 17: **Calibration Metrics.** A demonstration of how we measure calibration and selective classification. The model probabilities refer to the probabilities the model assigns to its prediction. For simplicity, the figure uses 2 bins for ECE computation, but we use 10 bins in practice.

models inform decision-making (e.g. resume screening), which we increasingly see for language technology as its scope broadens. For example, if a model is uncertain in its prediction, a system designer could intervene by having a human perform the task instead to avoid a potential error (i.e. selective classification). To concretize how uncertainty quantification is specifically useful in the context of language models, two examples include using model confidences/uncertainties to inform how to aggregate different prompts (Arora et al., 2022) and assemble prompt chains (Wu et al., 2022). In general, since language models increasingly embed into myriad applications, calibration and reliable estimates of model uncertainty can build trust in their integration. Figure 17 depicts how we measure calibration; see Appendix C.2 for more details.

Calibration (Murphy, 1973; Murphy & Winkler, 1977; DeGroot & Fienberg, 1983) is a widely studied property in the literature on uncertainty quantification: a model is calibrated if it assigns meaningful probabilities to its predictions. Concretely, if a well-calibrated model predicts that 1,000 sentences are toxic each with probability 0.7, then we expect around 700 of them to be toxic. To quantify calibration, we compute the expected calibration error (ECE; Naeini et al., 2015; Guo et al., 2017), which measures the difference between the model's predicted probability and the fraction of times the model is correct. By default, we use 10-bins with an equal number of probabilities per bin.

We also test the potential for *selective classification* (El-Yaniv & Wiener, 2010; Geifman & El-Yaniv, 2017): we evaluate the accuracy for the C-fraction of examples the model assigns the highest probability, where the model abstains for the remaining $1 - C$ examples. We report both the selection classification accuracy for $C = 0.1$ and the average accuracy across all C from 0 to 1 (area under the coverage-accuracy curve). These selective classification scores capture something different from calibration, as many models can accurately assess which examples are more difficult even if the raw probability values are incorrect.

4.5 Robustness

When deployed in practice, models are confronted with the complexities of the open world (e.g. typos) that cause most current systems to significantly degrade (Szegedy et al., 2014; Goodfellow et al., 2015; Jia & Liang, 2017; Belinkov & Bisk, 2018; Madry et al., 2018; Ribeiro et al., 2020; Santurkar et al., 2020; Tsipras, 2021; Dhole et al., 2021; Koh et al., 2021; Yang et al., 2022). Thus, in order to better capture the performance of these models in practice, we need to expand our evaluation beyond the exact instances contained in our scenarios (Jia & Liang, 2017; Goel et al., 2021; Dhole et al., 2021; Wang et al., 2021b).

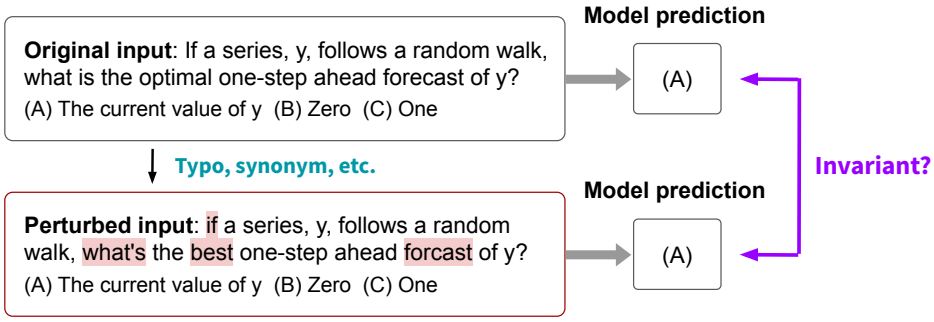

Figure 18: **Robustness perturbations.** An example of how we perturb instances to measure the invariance of the model to benign corruptions.

Towards this goal, we measure the *robustness* of different models by evaluating them on transformations of an instance. That is, given a set of transformations for a given instance, we measure the worst-case performance of a model across these transformations (Figure 18). Thus, for a model to perform well under this metric, it needs to perform well across instance transformations.

Specifically, we will focus on two notions of transformations—namely *invariance* and *equivariance*—described below (see Appendix C.3 for more details). Note that both of these capture the *local* robustness of a model, that is how robust the model is to transformations in the neighborhood of each instance. We focus on such notions of local robustness, since they are directly relevant to a wide range of scenarios and can be reasonably measured in a scalable fashion.

However, we emphasize that the other forms of robustness are important, but we find that they are comparatively more difficult to measure because of the lack of assumptions we make on the models we evaluate as well as the scale of our evaluation. Specifically, on one hand, measuring robustness to distribution or subpopulation shift (Oren et al., 2019; Santurkar et al., 2020; Goel et al., 2020; Koh et al., 2021) requires scenarios with special structure (i.e., explicit domain/subpopulation annotations) as well as information about the training data of the models. On the other hand, measuring adversarial robustness (Biggio et al., 2013; Szegedy et al., 2014) requires many *adaptive* queries to the model in order to approximat worst-case perturbations, which are not feasible in this evaluation (Wallace et al., 2019a; Morris et al., 2020). Finally, a recent line of work has explored interactive human-in-the-loop adversarial evaluation (Wallace et al., 2019b; Nie et al., 2020; Bartolo et al., 2020; Kiela et al., 2021), including work on red teaming models (Perez et al., 2022; Ganguli et al., 2022), which we believe is very relevant but difficult to scale for our purposes.

Invariance. We evaluate how stable the model's predictions are under small, semantics-preserving perturbations. This transformation/perturbation-based paradigm has been widely explored to study model robustness (e.g. Ribeiro et al., 2020; Goel et al., 2021; Wang et al., 2021a), with our implementation drawing significantly from `NL-Augmenter` (Dhole et al., 2021).[42] The goal is to understand whether corruptions that arise in real use-cases (e.g. typos) affect the performance of the model significantly. Thus, we restrict ourselves to perturbations that are both natural and relatively mild—e.g., capitalization, common misspellings—see Figure 18 for an illustration and see Appendix D.1 for the full description. Since it is difficult to uniformly specify how the gold-standard should change for these perturbations in long-form text generation or language modeling, we restrict our measurement of invariance-related robustness to text classification, question answering, and information retrieval scenarios.

Equivariance. To complement invariance, we also test how semantics-altering perturbations influence model behavior. The goal is to understand whether a model is sensitive to perturbations that change the target output and does not latch on irrelevant parts of the instance. Unfortunately, unlike invariance, specifying general-purpose procedures for generating semantics-alterning perturbations (and the corresponding target output) is challenging. Thus, we rely on *Contrast Sets* (Gardner et al., 2020), a resource which consists

[42] `https://github.com/GEM-benchmark/NL-Augmenter`

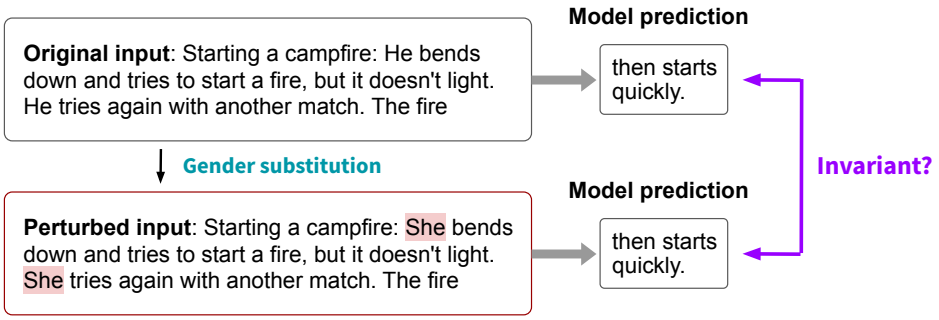

Figure 19: **Fairness Perturbations.** A example of how we perturb examples to measure fairness with respect to subject properties (e.g. the gender of the entities mentioned in the text).

of transformed versions of existing datasets (generated by the authors of the original datasets), aimed to test equivariance through counterfactually-augmented data (Kaushik et al., 2019). Since such contrast sets only exist for a few datasets, we use contrast sets when they are available (i.e. the **BoolQ** question answering scenario and the **IMDB** sentiment analysis scenario). Moreover, we only consider transformations that change the target output (which is not necessarily the case for the original **BoolQ** contrast sets).

4.6 Fairness

The disparate treatment and disparate impact (Barocas & Selbst, 2016) of machine learning is well-documented (Sweeney, 2013; Howard et al., 2017; Buolamwini & Gebru, 2018; Noble, 2018; Benjamin, 2019, *inter alia*), including in the context of language technologies (e.g. Koenecke et al., 2020). Centering fairness and equity as first-class aspects of evaluation is therefore essential to ensuring technology plays a positive role in social change (Friedman & Nissenbaum, 1996; Abebe et al., 2020; Bommasani et al., 2021, §5.1). We operationalize fairness measurement in two ways following Khani & Liang (2020): (i) *counterfactual fairness* (Dwork et al., 2012; Kusner et al., 2017) and (ii) statistical fairness or *performance disparities*. See Appendix C.4 for more details.

Counterfactual fairness. By counterfactual fairness, we refer to model behavior on *counterfactual* data that is generated by perturbing existing test examples (cf. Ma et al., 2021; Qian et al., 2022), akin to our approach for testing model robustness to invariances (§4.5: METRICS-ROBUSTNESS). These perturbations correspond either to social groups involving either (i) the *speaker* who produced the data (e.g. African American English) or (ii) the *subject* of the text who is mentioned within it. We consider several perturbations, which augment the original test instances with additional instances that substitute specific group-related terms with alternatives (see Figure 19). In Appendix D.2, we provide the specific terms and probabilities of substitution. Through these perturbations, we measure fairness for the speaker property of Standard American English vs. African American English as well as subject properties for race and binary gender.[43] Akin to our approach for robustness, we restrict our measurement of counterfactual fairness to text classification, question answering, and information retrieval scenarios to better ensure the validity of the perturbations.

Performance disparities. While perturbation-based methods for counterfactual fairness afford both control and scalability (to arbitrary scenarios), which facilitates evaluation across many scenarios, they are limited. Specifically, since the underlying distribution depends on one group's data (i.e. the group whose data is being perturbed), they fail to reflect unfairness when the data distributions across groups differ in more complex ways. Consequently, we measure *performance disparities* for scenarios where test instances are annotated with (pre-existing) group-level metadata by reporting how the accuracy on the subset of the test set corresponding to each group. Since these measurements depend on the availability of group-level metadata,[44] we cannot produce such measurements for most scenarios. However, across the benchmark, we

[43]Evaluation for some intersectional groups (Crenshaw, 1989) is straightforward given our approach, but left for future work.
[44]Future work may choose to explore automated methods for inferring groups, and the errors in such approaches, as a more scalable approach.